

Abdul Samad Gilal

Portfolio: abdulsgilal.com | abdulsgilal@gmail.com | linkedin.com/in/abdulsgilal/ | github.com/abdulsgilal579

Profile Introduction

Software Engineer with a Bachelor's in Computer Systems Engineering, experienced in building REST APIs, full-stack web applications, and production-grade AI systems. Proficient in Node.js, JavaScript, Python, and Redis, with hands-on experience deploying containerized services on cloud infrastructure. Eager to contribute to scalable payment and enterprise software solutions.

Education

Master's in Computer Science

Class of 2027

FAST NUCES, Karachi

Bachelor of Computer Systems Engineering

Jan 2021 – Dec 2024

Mehran University of Engineering & Technology, Jamshoro, Pakistan

Work Experience

AI/ML Engineer – Data Pulse Tech LLC, NASTP Karachi

Oct 2025 – Feb 2026

- Developed a computer vision pipeline using MediaPipe for hand tracking of baseball players, detecting pitch release points and throw mechanics
- Built and evaluated a RAG application using RAGAS to benchmark retrieval accuracy and response quality
- Deployed AI systems on Google Cloud Platform, managing infrastructure for production workloads

Research Activities

A Comparative Study of Chain-of-Thought Faithfulness in LLaMA-3.1 8B and LLaMA-4 Scout 17B Using Step Intervention Across Mathematics, Factual, and Commonsense Domains

2026

Manuscript in preparation – Investigates CoT faithfulness in two LLaMA models via step-intervention methodology across three reasoning domains.

Projects

Finance Tracker — Personal Finance Web App [GitHub]

- Built a full-stack personal finance application implementing the 50-30-20 budgeting rule, featuring a Node.js/Express REST API backend, React frontend, and persistent SQLite database via better-sqlite3
- Implemented multiple income sources, goal tracking with priority levels (High/Medium/Low), an Emergency Fund module with withdrawal tracking, and real-time spent/remaining indicators per budget category
- Node.js - Express - React - Vite - SQLite - Recharts - Lucide React

RAG Pipeline with Semantic Caching [GitHub]

- A high-performance RAG pipeline that extracts text from PDFs, stores vector embeddings in Qdrant, and uses Redis as a semantic cache to reduce redundant LLM calls lowering latency and API cost while using Groq for response generation.
- Redis - Qdrant - Hugging Face Transformers - PyMuPDF - Docker - Python - Groq LLM

VisionAsk — Real-Time Scene & System [GitHub]

- Built a real-time scene Q&A system using object detection, multi-object tracking, and an LLM backend — enabling natural language queries over live webcam feeds with grounded, accurate responses.
- Python - YOLOv8 - OpenCV - SORT (Kalman Filter) - Groq - Whisper

Magic AI Captions [GitHub]

- Built an AI-powered video captioning pipeline using OpenAI Whisper for speech recognition, burning captions directly into video via FFmpeg and uploading processed output to AWS S3 with downloadable links.
- OpenAI Whisper - FFmpeg - AWS S3 - FastAPI - Python - Docker

Tech Stack

Python, JavaScript (Node.js), Go · REST API Development · FastAPI, Flask, Express.js
React, Tailwind CSS · Redis (caching, in-memory datastore) · SQLite, PostgreSQL
AWS, Docker, Git · LLM APIs, RAG, Vector Databases, Computer Vision

Certifications

AWS Cloud • Docker • Generative AI Langchain • Machine Learning
Pandas • Version Control Git • Javascript